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Experiment 1 Implementation of a Single Layer Perceptron for Binary Classification

1. Objective

The objective of this experiment is to understand the concept of an artificial neuron and implement a Single Layer Perceptron from scratch for binary classification. Students will learn the perceptron learning algorithm, understand the importance of activation functions, visualize the learning process, and evaluate the classifier using a real-world dataset.

2. Background Theory

An Artificial Neuron is the fundamental building block of a neural network. It receives one or more input features, multiplies them by corresponding weights, adds a bias term, and applies an activation function to determine the output.

The Single Layer Perceptron is the earliest supervised learning algorithm capable of solving linearly separable binary classification problems.

Mathematical Formulation

Weighted Sum

$$z = \mathbf{w}^T \mathbf{x} + b$$

where

- $\mathbf{x}$: Input vector
- $\mathbf{w}$: Weight vector
- b : Bias
- z : Linear combination

Prediction

$$\hat{y} = f(z)$$

Weight Update Rule

$$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(y - \hat{y})\mathbf{x}$$

Bias Update

$$b_{new} = b_{old} + \eta(y - \hat{y})$$

where η denotes the learning rate.

3. Activation Functions

An activation function determines the output of a neuron based on the weighted sum of its inputs. It introduces non-linearity, enabling neural networks to learn complex patterns. Although the perceptron uses only the Step Activation Function, modern deep learning models employ differentiable activation functions for efficient optimization through backpropagation.

Step Activation Function

$$f(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

The Step Activation Function produces a binary output and is therefore suitable only for linearly separable binary classification problems.

Activation	Output Range	Typical Applications
Step	$\{0, 1\}$	Classical Perceptron
Sigmoid	$(0, 1)$	Binary Classification Output Layer
Tanh	$(-1, 1)$	Hidden Layers (Traditional Networks)
ReLU	$[0, \infty)$	Hidden Layers in CNNs and MLPs
Leaky ReLU	$(-\infty, \infty)$	Avoiding Dying ReLU
Softmax	$(0, 1)$	Multi-class Classification Output Layer

Note: This experiment uses only the Step Activation Function. The remaining activation functions will be studied in subsequent experiments involving Multilayer Perceptrons and Deep Neural Networks.

4. Dataset Description

Dataset: Banknote Authentication Dataset

Source: UCI Machine Learning Repository

Download Link:

<https://archive.ics.uci.edu/dataset/267/banknote+authentication>

Instances	1372
Features	4 Numerical Features
Classes	2
Missing Values	None
Task	Binary Classification

Feature Description

- Variance
- Skewness
- Curtosis
- Entropy

Target Class

- 0 – Authentic Banknote
- 1 – Forged Banknote

5. Experimental Procedure

Task 1: Dataset Exploration

- Load the dataset.
- Display the first five samples.
- Determine dataset dimensions.
- Identify missing values.
- Display descriptive statistics.

Expected Output

Dataset summary and statistical information.

Task 2: Exploratory Data Analysis

Generate

- Histograms
- Correlation Heatmap
- Scatter Plot
- Boxplots

Task 3: Data Preprocessing

- Normalize all numerical features.
- Split the dataset into Training (80%) and Testing (20%).

Task 4: Perceptron Implementation

Implement from scratch

- Weight Initialization
- Bias Initialization
- Step Activation Function
- Forward Propagation
- Perceptron Learning Rule

Task 5: Model Training

Display

- Epoch Number
- Misclassified Samples
- Updated Weights
- Updated Bias

Task 6: Model Evaluation

Compute

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

6. Source Code

The complete implementation of the Single Layer Perceptron, including data preprocessing, model training, testing, and result visualization, is provided in the accompanying Kaggle Notebook.

Kaggle Notebook:

<https://www.kaggle.com/code/kalpanabhaskar/single-layered-perceptron>

7. Experimental Results

Training Summary

Dataset Size	4 features x 1732 instances
Train/Test Split	80 - 20 ratio
Learning Rate	0.01 (Doesnt affect misclassified samples as step function is not differentiable)
Epochs	10 (6th Epoch results in least number of misclassified samples)
Final Weights	[-1.0268 -1.2769 -1.1338 -0.0096]
Final Bias	-0.6
Accuracy	0.9964
Precision	0.9919
Recall	1.0000
F1-score	0.9959

Epoch-wise Learning

Epoch	Errors	Weight 1	Weight 2	Bias
1	59	-0.6803	-0.9381	-0.3
2	25	-0.7754	-1.0421	-0.4
3	20	-0.8021	-1.1266	-0.4
4	19	-0.8828	-1.1959	-0.5
5	20	-0.9909	-1.2951	-0.5
6	17	-1.0268	-1.2769	-0.6
7	24	-1.0813	-1.3928	-0.6
8	22	-1.1702	-1.3568	-0.6
9	20	-1.1755	-1.4696	-0.6
10	20	-1.3069	-1.5412	-0.6

8. Generated Plots with Interpretation

1. Feature Histograms

Figure 1: Feature Histograms

- The histograms show the distribution of the four numerical features. Variance and Skewness exhibit a wider spread, while Kurtosis and Entropy show skewed distributions.
- The differences in feature scales and distributions justify the use of feature normalization before training the perceptron.

2. Correlation Heatmap

Figure 2: Correlation Heatmap

- The heatmap illustrates the pairwise correlation between all features and the target class, highlighting both positive and negative relationships.
- Some features exhibit moderate to strong correlations with the class label, indicating their usefulness for binary classification.

3. Scatter Plot

Figure 3: Scatter Plot

- The scatter plot visualizes the distribution of authentic and forged banknotes using two selected features.
- Although a few samples overlap, the classes appear reasonably linearly separable, making the dataset suitable for a Single Layer Perceptron.

4. Training Error vs Epoch

Figure 4: Training Error vs Epoch

- The number of misclassified samples decreases over successive epochs, indicating that the perceptron successfully learns from the training data.
- The error stabilizes after a few epochs, demonstrating convergence of the learning algorithm.

5. Weight Evolution

Figure 5: Weight Evolution

- Each weight changes during training as the perceptron updates its parameters to reduce classification errors.
- The gradual stabilization of the weights indicates that the model has converged to a suitable solution.

6. Bias Evolution

Figure 6: Bias Evolution

- The bias is adjusted throughout training to shift the decision boundary appropriately.
- The bias eventually stabilizes, indicating that further updates provide little improvement in classification.

7. Confusion Matrix

Figure 7: Confusion Matrix

- The confusion matrix summarizes the classification performance by showing the numbers of true positives, true negatives, false positives, and false negatives.
- A high number of correctly classified samples and relatively few misclassifications indicate good predictive performance of the perceptron.

8. Learning Rate Comparison

Figure 8: Learning Rate Comparison

- Different learning rates influence the speed at which the perceptron converges during training.
- A learning rate of 0.1 converges faster than 0.01 and 0.001 for this dataset, while excessively large learning rates may lead to unstable learning in general.

9. Decision Boundary (Optional)

Figure 9: Decision Boundary

- The decision boundary illustrates the linear separation learned by the perceptron using two selected features for visualization.
- Most samples are correctly separated into authentic and forged classes, while overlapping points indicate the limitations of representing a four-dimensional classifier in two dimensions.

9. Discussion

- The Single Layer Perceptron was successfully implemented from scratch using the perceptron learning algorithm with weight updates and bias adjustment.
- Exploratory Data Analysis showed that the Banknote Authentication dataset contains numerical features with different ranges and distributions. Feature normalization was therefore applied to improve training stability and convergence.
- The boxplot analysis identified outliers in features such as Kurtosis and Entropy. These values were retained because they represent valid observations in the dataset rather than incorrect data points.
- The perceptron showed a reduction in misclassified samples over successive epochs, demonstrating that the model learned an effective linear decision boundary.
- The choice of learning rate affected the convergence speed of the model. A higher learning rate resulted in faster updates, while smaller learning rates required more epochs to achieve similar performance.
- The evaluation metrics including accuracy, precision, recall, and F1-score indicated the effectiveness of the trained perceptron in classifying authentic and forged banknotes.
- The confusion matrix provided insights into correctly classified samples and misclassification patterns, helping to understand the strengths and limitations of the classifier.
- The decision boundary visualization showed that the perceptron can separate the classes approximately linearly. However, the visualization is limited because the original dataset contains four features, while the plot represents only two selected features.
- Compared with modern deep learning models, the perceptron has limitations because it can only solve linearly separable problems and cannot learn complex non-linear relationships.

10. Conclusion

- The experiment provided an understanding of the fundamental working principle of an artificial neuron and the Single Layer Perceptron model.
- A perceptron classifier was successfully implemented from scratch using the step activation function and perceptron learning rule.
- Data preprocessing, including feature normalization and train-test splitting, improved the reliability and convergence of the model.
- The model successfully learned from the Banknote Authentication dataset and achieved effective binary classification performance.
- The effect of learning rate on model convergence was analyzed, demonstrating the importance of selecting appropriate hyperparameters.
- The experiment highlighted the advantages and limitations of the perceptron algorithm, forming the foundation for understanding advanced neural networks and deep learning models.

- The study demonstrates that while a Single Layer Perceptron is simple and efficient for linearly separable problems, multilayer neural networks are required for solving complex non-linear classification tasks.

11. References

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